A

Major Project Report

on

Document Image Enhancement using Vision Transformers

Submitted in partial fulfilment of the requirement for the award of the degree of

BACHELOR OF TECHNOLOGY
in

COMPUTER SCIENCE AND ENGINEERING

Submitted by

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CERTIFICATE

This is to certify that the Major Project report on "**Document Image Enhancement using Vision Transformers**" is a bonafide work carried out by **Vedashri Vitthal Gore (20R91A05G2), Sabavat Swapna (20R91A05E0)** and **Ponnala Pranay (20R91A05C6)** in the partial fulfilment for the award of B.Tech. degree in Computer Science and Engineering, Teegala Krishna Reddy Engineering College (TKREC-R9), Hyderabad, affiliated to Jawaharlal Nehru Technological University, Hyderabad under our guidance and supervision.

The result of investigation enclosed in this report has been verified and found satisfactory. The results embodied in the project work have not been submitted to any other University for the award of any degree.

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DECLARATION

We hereby declare that the Major Project report entitled **"Document Image Enhancement using Vision Transformers"** is done under the guidance of **Dr. CH. V. Phani Krishna, Professor,** Department of Computer Science and Engineering, Teegala Krishna Reddy Engineering College, is submitted in partial fulfilment of the requirements for the award of the degree of **Bachelor of Technology in Computer Science and Engineering** from **Jawaharlal Nehru Technological University, Hyderabad**.

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ABSTRACT

Document images can undergo numerous degradation scenarios, causing difficulties in recognition and processing. In the digital era, it is crucial to remove noise from these images to ensure their proper utilization. To tackle this issue, a new encoder-decoder architecture based on vision transformers has been developed to improve both machine-printed and handwritten document images seamlessly. The encoder works directly on pixel patches along with their positional data, eliminating the need for convolutional layers, and the decoder recreates a clean image from the encoded patches. Experimental results indicate the superiority of this model over existing methods on various DIBCO benchmarks.

Keywords: Degraded images, Encoder-Decoder Architecture, DIBCO

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1. INTRODUCTION

1.1 Problem Statement

The main problem for this paper focuses on the degradation scenarios that affect the document images, leading to challenges in recognition and processing. Degradation scenarios can be blurriness, oil stains, water marks, pen strokes, artefacts, and uneven illumination. The specific issues faced include the recognition difficulties caused by degraded images and necessity to denoise them for proper usage of these images. The primary challenge addressed by the project is the need to enhance both the handwritten and machine-printed document images by developing a new encoder-decoder architecture based on vision transformers.

1.2 Description

Preserving the legibility and quality of document images, especially historical ones, is a critical focus within Document Image Analysis and Recognition(DIAR) research. Historical documents often hold valuable information spanning centuries and decades. However, these documents are susceptible to various formsof degradation such as smears, stains, pen strokes, and uneven illumination, whichcan hinder their preservation.

The presence of such distortions can significantly impact downstream tasksin information processing, including segmentation, Optical Character Recognition (OCR), information spotting, and layout analysis. To address these challenges, a robust preprocessing step is essential to denoise and reconstruct degraded images into high-quality, clean versions. Document Image Enhancement (DIE) plays a crucial role in restoring the quality of degraded document samples to produce clear,uniform images.

Recent advancements in Convolutional Neural Network (CNN)-based approaches have revolutionized DIE tasks such as binarization, deblurring, shadow removal, and watermark removal. While CNNs have shown improved performanceover traditional methods, they have limitations. CNNs may not be optimal for restoring different regions of a document image due to their grid-based operation and limited ability to capture long-range dependencies.

In response to these limitations, Vision Transformers (ViTs) have emergedas a promising alternative. ViTs split images into fixed-size patches, enabling the capture of both local and global spatial dependencies. By leveraging the self- attention mechanism inspired by transformers in Natural Language Processing (NLP), ViTs excel at capturing global interactions and contextual features inimages.

Our proposed Document Image Enhancement using Vision Transformer leverages ViTs in an encoder-decoder framework, eliminating the need for CNNs.This aims to restore and enhance degraded document images efficiently. By incorporating ViTs and self-attention mechanisms, it can recover missing or degraded patches by reasoning globally between neighbouring patches.

The innovative approach of this project introduces a transformer-based model for document image enhancement, showcasing the power of ViTs. Experimental results demonstrate that this project achieves state-of-the-art performance in document binarization tasks, including machine-printed and handwritten documents.

2. LITERATURE SURVEY

1. Transformers in Vision: A Survey - arXiv

It discusses image restoration Transformer models for various tasks like image super-resolution, image enhancement, and colorization. It compares advantages and limitations of popular techniques in computer vision. It highlights the applications of transformers in image processing tasks and provides insights into future researchdirections.

2. An Overview of Vision Transformers for Image Processing: A Survey - IJACSA

It focuses on the success of vision transformer-based models in artificial intelligence, particularly in computer vision. It emphasizes the potential applications ofvision transformers for student verification in university systems. It discusses the importance of accurately verifying student identities for academic success in online learning environments.

3. A Survey on Visual Transformer - arXiv

It provides a complete overview of transformer-based models in computer vision, covering object detection, image classification, and video processing. It discusses the applications of transformers in high/mid-level vision, low-level vision, and videoprocessing tasks. It highlights the performance of transformer models in various visualtasks and their potential for improving a wide range of visual applications.

4. Vision Transformers in Image Restoration: A Survey - MDPI

It surveys the impact of Vision Transformers in image restoration tasks like imagesuper-resolution, denoising, and deblurring. It compares the efficiency metrics of ViT-based models on different benchmarks and datasets for image restoration. It discusses the challenges and future work in using Vision Transformers for image restoration tasks.

3. SYSTEM ANALYSIS

3.1 Existing System

Convolutional Neural Networks (CNNs) are commonly used for improving document images. They work on regular grids and using the same filter for differentparts of a damaged document may not be the best choice. They struggle to capture high-level dependencies. Transformers offers new solutions to address CNN limitations.

Drawbacks:

1. Grid-Based Operation
2. Limited Capture of High-Level Dependencies
3. Task-Specific Filters
4. Less Effective for Long-Range Dependencies

3.2 Proposed System

In this proposed system, a simple Document Image Enhancement Transformer is used. It is an end-to-end approach that effectively restores and improves degraded document images. It utilizes Vision Transformers in an encoder-decoder framework. Document Image Binarization Competition (DIBCO) shows that this achieves state-of-the-art results for both machine-printed and handwrittendegraded document images

Advantages:

1. Transformer-Based Architecture
2. Encoder-Decoder Framework
3. Flexibility and Simplicity

3.3 System Requirements

3.3.1 Software Requirements

- Coding Language : Python
- Operating System : Windows
- Front-End : HTML, CSS

3.3.2 Hardware Requirements

- System : Intel Core i5
- Hard Disk : 80 GB
- RAM : 8 GB

3.4 Feasibility Study

In this phase, the project's feasibility is assessed, and a business proposal is presented with a preliminary project plan and cost estimates. The feasibility study of the proposed system is conducted during system analysis to ensure it does not impose a financial burden on the company. Understanding the major systemrequirements is crucial for feasibility analysis, with three key considerations:

- Economical Feasibility
- Technical Feasibility
- Social Feasibility

3.4.1 Economical Feasibility

This evaluation examines the economic impact of the system on the organization. Given limited research and development funds, expenditures must be justified. The system was developed within budget constraints by leveraging freely available technologies and only purchasing customized products.

3.4.2 Technical Feasibility

This assessment focuses on the technical requirements of the system. It is essential that the system does not strain existing technical resources, avoiding excessive demands on the client. The developed system should have modest technical requirements, necessitating minimal or no changes for implementation.

3.4.3 Social Feasibility

This aspect evaluates the system's acceptance by users, including user training to ensure efficient system utilization. Users should perceive the systemas a necessity rather than a threat. User acceptance depends on effective education methods and user familiarization. Building user confidence encourages constructive criticism, valuable as users are the ultimate beneficiaries of the system.

4. SYSTEM DESIGN

System Design, also known as top-level design, focuses on determining the system's modules, their specifications, and their interconnections to achieve the intended outcomes. Upon completion of the system design phase, key decisions are made regarding major data structures, file formats, output formats, the primary modules within the system, and their respective specifications. Software design forms the bedrock of the software engineering process, transcending development paradigms andapplication domains. It serves as the initial step in crafting any engineered product or system, with the primary goal of creating a model or representation of the entity to be constructed. Following the delineation and analysis of system requirements, system design emerges as the foremost technical activity among design, code, and test, crucialfor software construction and validation. The essence of software design lies in fostering quality within software development, providing representations that can be evaluated for their quality and serving as the conduct through which a customer's visionis accurately translated into a final software product or system.

The design phase marks a pivotal shift from the problem domain to the solution domain, guiding the transition from understanding what is needed to devising how thoseneeds can be met. System design, also known as top-level design, outlines the system's modules, their specifications, interactions, data structures, file formats, output formats,and major modules, setting the foundation for the desired outcomes. In contrast, Detailed Design delves into the internal logic of each module specified in system design, articulating data details in a high-level design language independent of the eventual implementation language. Design is fundamentally about identifying software components, defining relationships among them, structuring software, and providing ablueprint for subsequent phases.

4.1 Software Development Life Cycle (SDLC)

SDLC is a process used to plan, design, develop, test, deploy, and maintainthe software applications. It is used by different software development organizations. SDLC provides a systematic, structured, and organized approach to software development. The system development life cycle adopted in this paper isAgile Methodology. The reason to select this methodology is due to its adaptability, iterative nature, and responsiveness to evolving requirements.

4.1.1. Agile Methodology

Agile Software Development is an incremental and iterative approach to software development. It emphasizes flexibility, rapid iteration, and continuous value delivery. Unlike other software development methods, Agile does not offera solution, instead, it offers various frameworks customized to specific industries,project types and team size.

Fig. 4.1 Agile Methodology

Advantages of Agile Methodology:

- Higher quality products
- Increased customer satisfaction
- Adaptability to change
- Faster time-to-market
- Enhanced collaboration and communication
- Risk Management

4.2 System Architecture

The proposed system architecture consists of the encoder-decoder framework which works on the basis of Vision Transformers.

1. **Input Image:** Degraded Image is taken as an input, it can be handwritten or printed.

2. **Image Patches:** The input image is divided into smaller pixel patches; each patch contains a subset of pixels from the original image.

3. **Flattened Pixel Patches:** The patches are then divided into a flattened pixel patch which represents the data in one-dimensional format, making the process further easy.

4. **Transformer Encoder:** This uses the Vision Transformers architecture and encodes the data from the flattened pixel patches using the positional information.

5. **Transformer Decoder:** This decodes the encoded data by reconstructing meaningful patterns, and uses the self-attention mechanisms for context-aware decoding.

6. **Combined Pixel Patches:** The decoded data is constructed into the combined pixel patch which represents the segments of the enhanced image.

7. **Enhanced Output:** This assembles all the combined pixel patches to form a complete enhanced image by improved quality, visual appeal, color correction or other features.

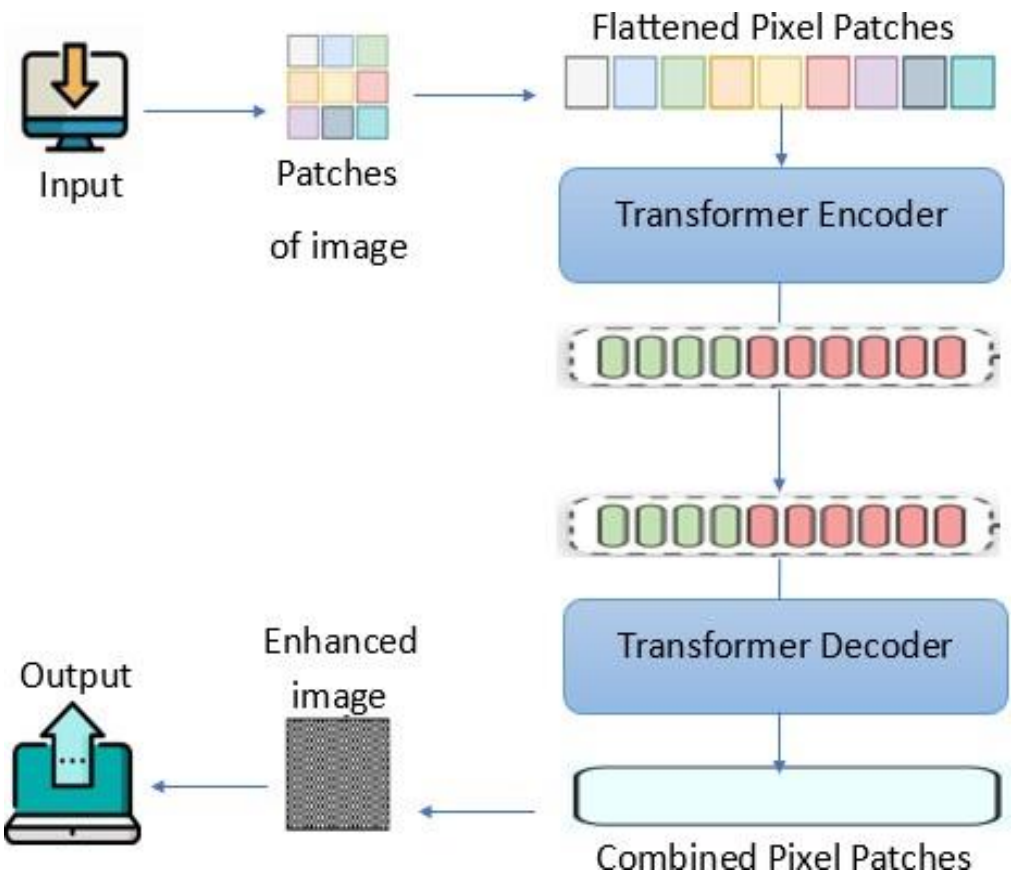

Fig. 4.2 Proposed System Architecture

4.3 Unified Modeling Languages(UML) Diagrams

Unified Modeling Language is a graphical language used for visualizing, constructing, and documenting the artifacts. UML Diagrams play a crucial role insoftware development and used graphical notations to express software project designs.

The UML Diagrams used in this paper are:

i. Class Diagram

ii. Use Case Diagram

iii. Sequence Diagram

iv. Activity Diagram

v. Deployment Diagram

i. Class Diagram

A class diagram is a fundamental component of the Unified Modeling Language (UML) used in software engineering. It provides a static view of an application by illustrating the structure of the system in terms of classes, their attributes, methods, and relationships among them.

Components of a Class Diagram:

1. **Class Names:** The top part of the diagram specifies the names of the classes.
2. **Attributes:** The middle section lists the attributes associated with each class.
3. **Methods (Operations):** The bottom section outlines the methods that a specific class can perform.

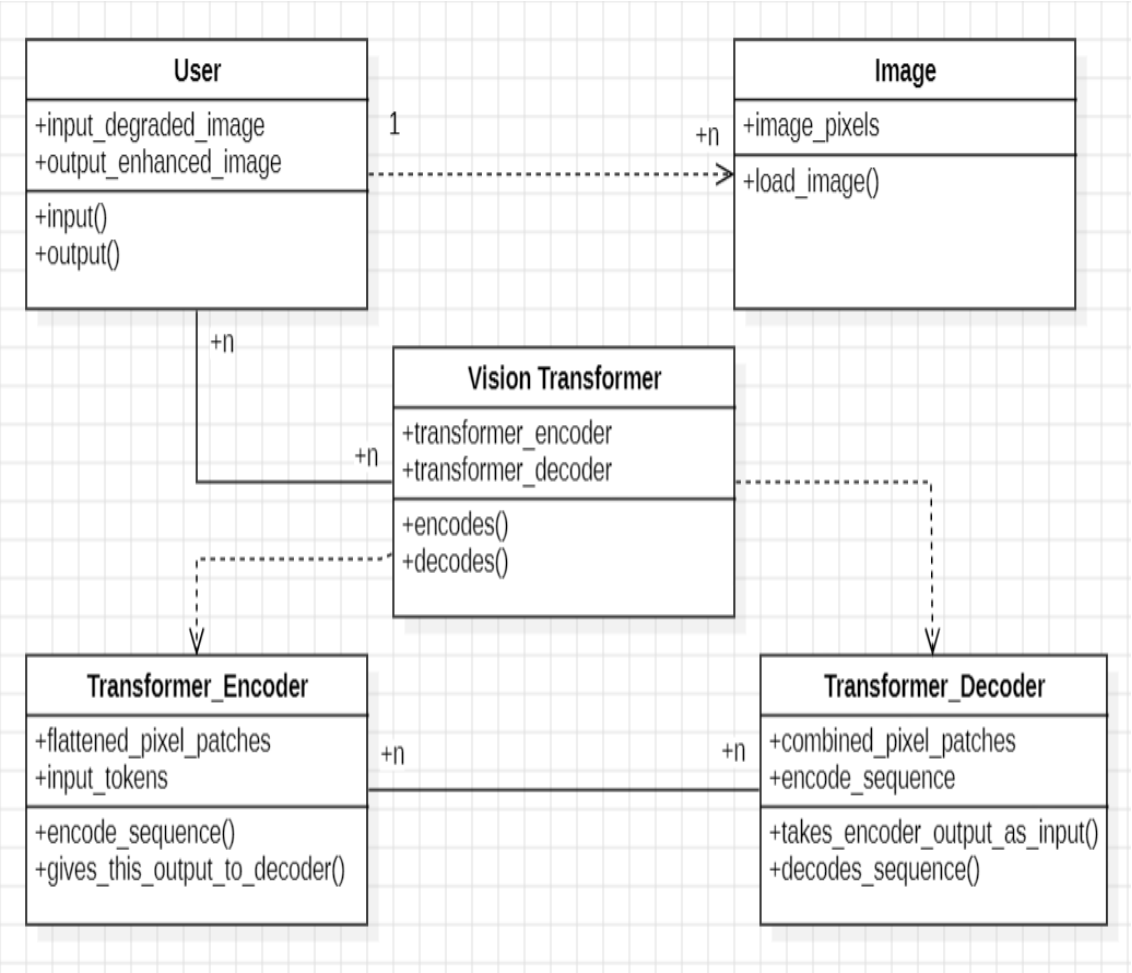

Fig. 4.3 Class Diagram

ii. Use Case Diagram

A use case diagram provides a visual representation of the various ways and scenarios in which a system can be used. It illustrates how users interact witha specific system, whether it is a website, an app, or any other software application.

Components of a Use Case Diagram:

1. **Use Cases:** These represent specific tasks, services, or functions that thesystem must perform. It encapsulates the system's functionality.

2. **Actors:** Actors are the external entities that interact with the system andtriggers the use cases.

3. **Relationships:** This shows how the actors and use cases are related.

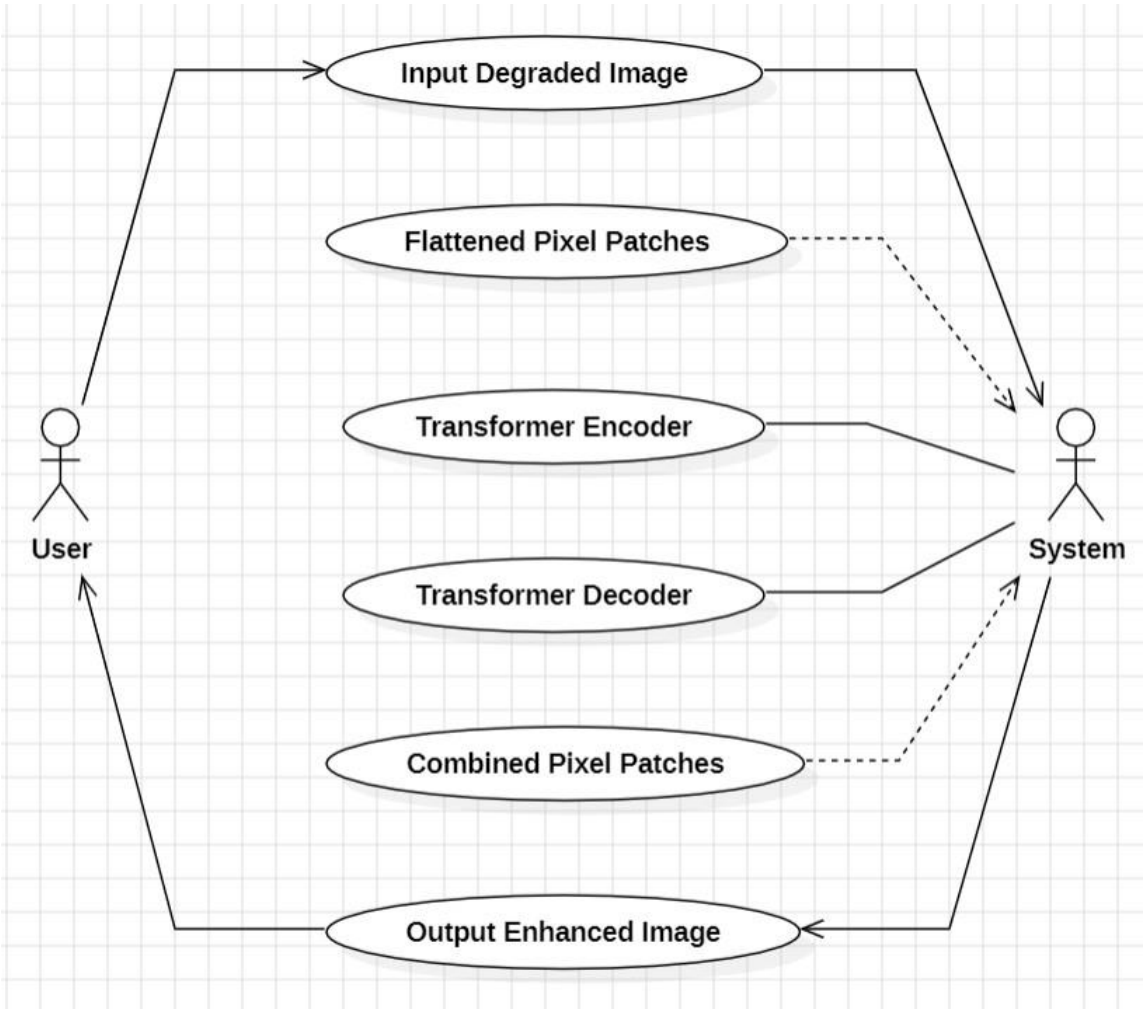

Fig. 4.4 Use Case Diagram

iii. Sequence Diagram

A sequence diagram provides a visual representation of how messages flowwithin a system. It is also known as an event diagram.

Components of a sequence diagram:

1. **Lifeline**: Represented by a vertical bar, the lifeline shows the existence of anobject or participant.

2. **Message Flow**: A vertical dotted line extends across the bottom of thediagram, indicating the flow of messages between lifelines.

3. **Iterations and Branching**: Sequence diagrams incorporate loops (iterations)and conditional branches, capturing the system's behavior comprehensively.

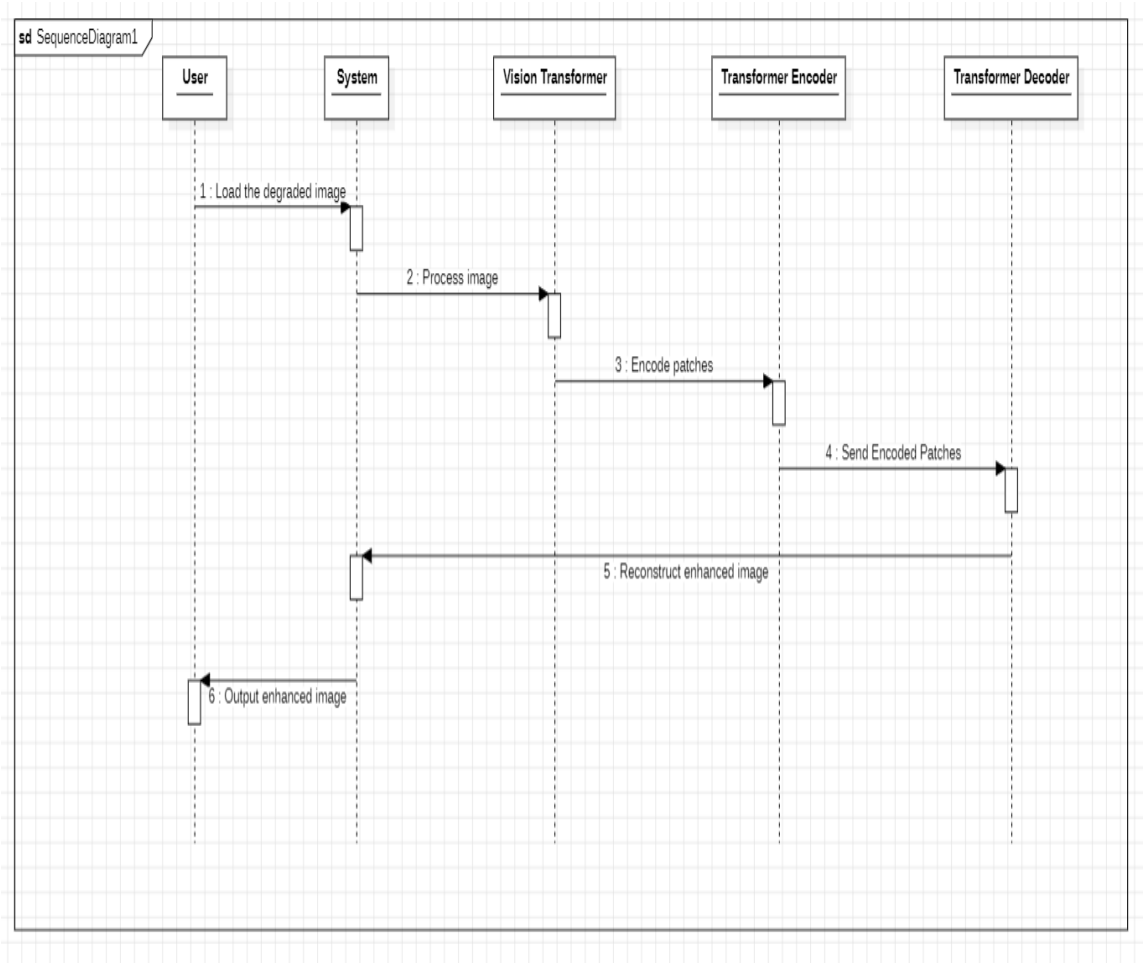

Fig. 4.5 Sequence Diagram

iv. Activity Diagram

An Activity diagram in the Unified Modelling Language (UML) serves asa tool for visualizing the flow of control within a system. It mainly focuses on thedynamic behavior of the system.

Components of an activity diagram:

1. **Activities**: These categorize behavior into one or more actions. An activity islike a network of nodes connected by edges, representing the flow of execution.
2. **Control Flow**: The control flow within an activity is depicted by control nodesand object nodes. These nodes represent the objects involved in the activity.
3. **Swim lanes**: Swim lanes cluster related activities in columns or rows.They enhance modularity in the diagram, although their use is optional.
4. **Forks**: Fork nodes generate concurrent flow within the activity. They have oneinward edge and several outward edges.

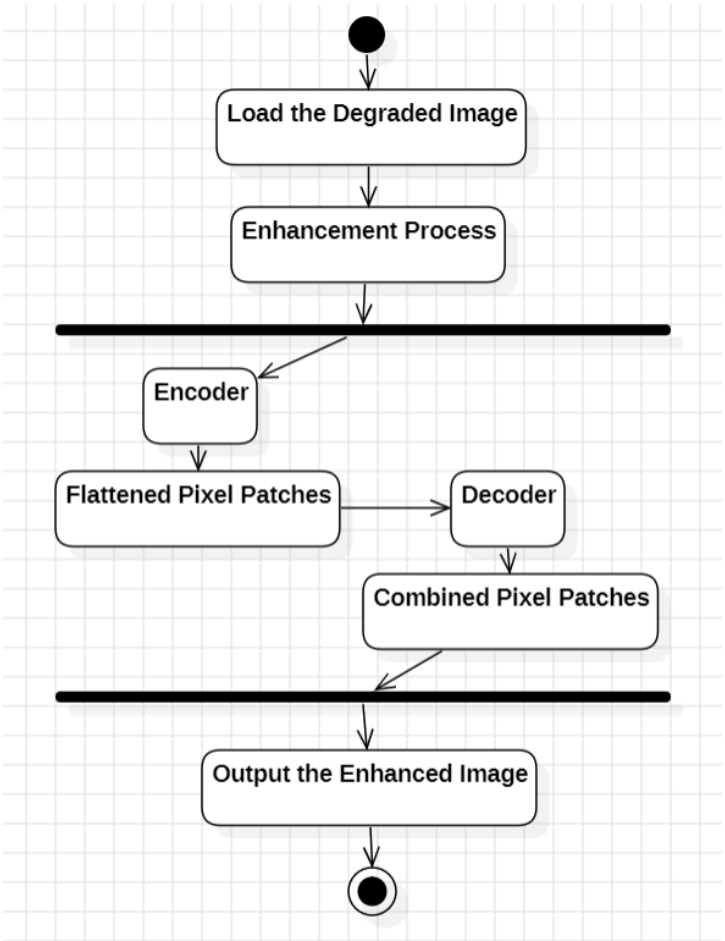

Fig. 4.6 Activity Diagram

v. Deployment Diagram

A Deployment Diagram in UML represents how software components arephysically deployed onto hardware nodes.

Components of Deployment Diagram:

1. **Nodes**: Nodes represent the physical hardware entities (servers, workstations,etc.).
2. **Components**: These represent the software modules or artifacts deployed onnodes.
3. **Artifacts**: In this, physical files (executables, databases) deployed on nodes.

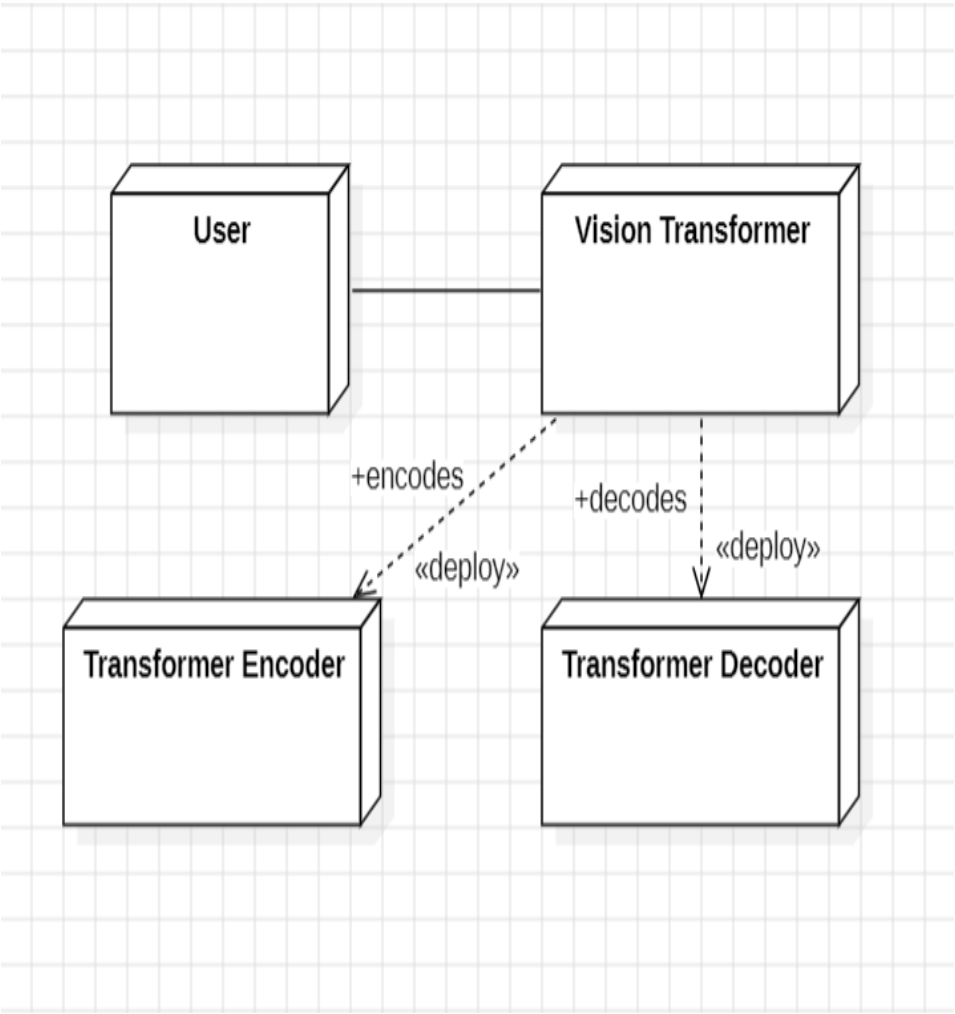

Fig. 4.7 Deployment Diagram

5. IMPLEMENTATION

5.1. Modules

In this paper, there are 2 modules based on the architecture of Vision Transformers.

5.1.1. Encoder

Within the encoding phase, the initial step involves partitioning an image intoa collection of patches. These patches are then transformed into tokens, incorporating their positional data. Subsequently, a series of transformer blocks is utilized to convert these tokens into the encoded latent representation. These blocksadhere to a structure similar to, comprising layers of multi-headed self-attention and multi-layered perceptron (MLP). Each block is prefaced by a LayerNorm (LN) andsucceeded by a residual connection. The dimensions of the patches' embedding andthe quantity of transformer blocks are determined based on the model's size.

a. Initial encoding stage divides image into patches

b. Patches embedded to create tokens with positional information

c. Transformer blocks used to map tokens into encoded latent representation

d. Blocks follow structure described in reference: multi-headed self-attention and multi-layered perceptron layers

5.1.2. Decoder

The decoder segment comprises a sequence of transformer blocks (equivalent in number to the encoder blocks) that receive the outputted tokens from the encoderas input. These tokens progress through the transformer decoder blocks and are thenprojected using a linear layer to generate the desired pixel values. This configuration ensures that each output element corresponds to a vector representinga flattened patch in the resultant image. Ground truth pixel values are derived by segmenting the ground truth (GT) clean image into patches and flattening them intovectors.

a. Decoder component consists of transformer blocks, matching encoder's number

b. Takes encoder's token sequence as input

c. Tokens processed within transformer decoder blocks

d. Projected using linear layer to get pixel values

e. Each output element represents flattened patch in resulting image

5.2. Development Tools used

5.2.1 Python

Python is used as a widely adopted high-level programming language with a broad applicability. Employed extensively in Web Development, Machine Learning, and various cutting-edge technologies within the Software Industry, the latest version, Python 3, serves as the language of choice.

Fig. 5.1 Python Logo

One of Python's notable strengths lies in its extensive standard library, which proves invaluable for diverse applications, including:

- Machine Learning
- Web scraping and frameworks
- Multimedia
- Test frameworks
- GUI Applications
- Image processing
- Speech Recognition

5.2.2 Visual Studio Code

Visual Studio Code (VS Code) is a lightweight and useful source code editor. It offers robust language support, including features like IntelliSense, a built-in debugger, and extensive customization options. VS Code provides a seamless coding experience, making it a popular choice for web development andmany other fields.

Fig. 5.2 Visual Studio Code Logo

Here are the key features of Visual Studio Code (VS Code):

- Lightweight and Fast
- Language Support
- IntelliSense and Code Navigation
- Built-in Debugger
- Customization
- Web Technologies
- Robust Architecture

5.2.3 PyTorch

PyTorch is a flexible and powerful deep learning framework that has gainedpopularity for its dynamic computation graph, making it well-suited for research and experimentation. Unlike static graph frameworks, PyTorch constructs graphs dynamically during execution, allowing for easier debugging, dynamic control flow, and custom gradients.

Its core data structure is the tensor, like NumPy arrays, and the autograd module automatically computes gradients for backpropagation. PyTorch provides a rich ecosystem of modules and libraries for building neural networks, including pre-defined layers, loss functions, and optimizers. Researchersand practitioners appreciate its compatibility with the broader SciPy stack and its interface that emulates MATLAB. To install PyTorch, you can use Conda or pip, and the community actively contributes through discussions, tutorials, and cutting-edge research. In summary, PyTorch empowers developers to create and experiment with neural networks effectively.

Fig. 5.3 Pytorch Logo

Here are some key features of Pytorch:

- Tensor Computation
- Automatic Differentiation
- Deep Neural Networks
- TorchServe for Model Deployment
- Distributed Training
- Mobile Deployment (Experimental)
- Robust Ecosystem

5.3. Libraries used

5.3.1 Numpy

Numpy serves as a versatile array-processing library, offering a powerful multidimensional array structure and a range of tools for manipulating these arrays. This essential package forms the backbone of scientific computing in Python, enabling efficient handling of multidimensional data. Additionally, Numpy facilitates the creation of custom data types, enabling seamless integrationwith diverse databases.

5.3.2 Pandas

Pandas, an open-source Python library, offers a robust set of tools for high-performance data manipulation and analysis through its advanced data structures. While Python traditionally focused on data preparation and cleansing, Pandas revolutionized data analysis by providing a comprehensive solution. With Pandas,users can seamlessly execute five key data processing and analysis steps—loading, preparing, manipulating, modeling, and analyzing data—regardless of its source. Python combined with Pandas finds extensive applications across various sectors such as academia, finance, economics, statistics, analytics, and more.

5.3.3 Scikit-learn

Scikit-learn offers a variety of supervised and unsupervised learningalgorithms through a uniform interface in Python. Licensed under a permissive simplified BSD license, it is widely available across numerous Linux distributions, promoting its adoption in both academic and commercial settings.

5.3.4 MatPlotLib

Matplotlib functions as a foundational graph plotting library in Python, offering a versatile visualization tool for generating a wide array of plots and charts. Developed by John D. Hunter, Matplotlib is an open-source resource accessible to all users. While primarily implemented in Python, Matplotlib incorporates elements of C, Objective-C, and JavaScript to ensure compatibility across different platforms.

5.4. Technologies used

5.4.1 Vision Transformers

A Vision Transformer (ViT) is a model designed for image classification tasks, revolutionizing traditional computer vision. Unlike conventionalconvolutional neural networks (CNNs), which operate on fixed-size image grids, ViTs break down an input image into smaller patches. Each patch is serialized intoa vector and mapped to a lower dimension using a single matrix multiplication. Inspired by the success of Transformers in natural language processing, ViTs have shown remarkable promise in various computer vision tasks, including image classification. These models employ a Transformer-like architecture over image patches, making them a powerful alternative for visual recognition tasks. The key concepts include "attention" and "multi-head attention," which play a crucial role in how vision transformers work. By enhancing relevant parts of the image while mimicking cognitive attention, ViTs learn to focus on important input features.

A few key features of Vision Transformers are:

- Efficiency
- Robust Feature Extraction
- Task-Specific Adaptability
- Interpretable Attention Maps
- Transfer Learning
- Spatial Hierarchies
- Positional Encodings
- Hybrid Architectures

5.4.2 DIBCO Dataset

The DIBCO (Document Image Binarization Contest) dataset is an international competition focused on document image binarization. It has been organized in conjunction with the ICDAR (International Conference on Document Analysis andRecognition) conferences. The primary objective of this contest is to identify advancements in binarization techniques for both machine-printed and handwrittendocument images. Participants evaluate their methods using performance measures aligned with document image analysis and recognition requirements. Over the years, DIBCO has seen different variants, including H-DIBCO (focused on handwritten documents). Researchers have tackled challenges related to noise, varying lighting conditions, and diverse document types. Binarization is a key preprocessing step for optical character recognition (OCR), and document indexing. The DIBCO dataset contributes to advancements in these areas. For thoseinterested in exploring the dataset and participating in the contest, DIBCO offers avaluable resource for advancing document image binarization techniques.

6. CODING

- **main.py:**

```
from typing import Any from thinc import configimport torch
from vit_pytorch import ViT


from models.binae import BINMODEL import torchvision.transforms as transformsimport
numpy as np
import torch.optim as optim from einops import rearrangeimport loadData2 as loadData
import utils as utils
from config import Configsimport os
import cv2 import glob
import matplotlib.pyplot as plt import matplotlib.image as mpimgcfg = Configs().parse()
FLIPPED = False THRESHOLD = 0.5
SPLITSIZE = cfg.split_size SETTING = cfg.vit_model_size
TPS = cfg.vit_patch_size
batch_size = cfg.batch_size

experiment = SETTING +'_'+ str(SPLITSIZE)+'_' + str(TPS)patch_size = TPS
image_size = (SPLITSIZE,SPLITSIZE)MASKINGRATIO = 0.5 VIS_RESULTS = True
TEST_DIBCO = cfg.testing_dataset

device = torch.device('cuda' if torch.cuda.is_available() else 'cpu')count_psnr =
utils.count_psnr
imvisualize = utils.imvisualize load_data_func = loadData.loadData_setsif SETTING ==
'base':
ENCODERLAYERS = 6

ENCODERHEADS = 8

ENCODERDIM = 768

if SETTING == 'small': ENCODERLAYERS = 3
ENCODERHEADS = 4
```

```
ENCODERDIM = 512

if SETTING == 'large': ENCODERLAYERS = 12
ENCODERHEADS = 16

ENCODERDIM = 1024

best_psnr = 0

best_epoch = 0
def sort_batch(batch): n_batch = len(batch)train_index = [] train_in = [] train_out = []
for i in range(n_batch):

idx, img, gt_img = batch[i]train_index.append(idx) train_in.append(img)
train_out.append(gt_img)
train_index = np.array(train_index)

train_in = np.array(train_in, dtype='float32') train_out = np.array(train_out, dtype='float32')
train_in = torch.from_numpy(train_in) train_out = torch.from_numpy(train_out) return
train_index, train_in, train_out
def test_data_loader():

_, _, data_test = load_data_func(flipped=FLIPPED)

test_loader = torch.utils.data.DataLoader(data_test, collate_fn=sort_batch,
batch_size=batch_size, shuffle=False, num_workers=2, pin_memory=True)

return  test_loader test_loader = test_data_loader()v = ViT(
image_size = image_size,patch_size = patch_size,

num_classes = 1000, dim = ENCODERDIM,
depth = ENCODERLAYERS,heads = ENCODERHEADS,
mlp_dim = 2048

)

model = BINMODEL(encoder = v,
```

```
masking_ratio = MASKINGRATIO,   #_________doesnt matter for binarizationdecoder_dim
= ENCODERDIM,
decoder_depth = ENCODERLAYERS,decoder_heads = ENCODERHEADS
)

model = model.to(device)

optimizer = optim.AdamW(model.parameters(),lr=1.5e-4, betas=(0.9, 0.95), eps=1e-08,
weight_decay=0.05, amsgrad=False)

def visualize(epoch):losses = 0
for i, (test_index, test_in, test_out) in enumerate(test_loader):# inputs, labels = data
bs = len(test_in)

inputs = test_in.to(device) outputs = test_out.to(device)with torch.no_grad():
loss,_, pred_pixel_values = model(inputs,outputs)

rec_patches = pred_pixel_values

rec_images = rearrange(rec_patches, 'b (h w) (p1 p2 c) -> b c (h p1) (w p2)',p1 = patch_size,
p2 = patch_size,  h=image_size[0]//patch_size)

for j in range (0,bs):

imvisualize(inputs[j].cpu(),outputs[j].cpu(),rec_images[j].cpu(),test_index[j],epoch,e
xperiment)

losses += loss.item()

print('valid loss: ', losses / len(test_loader))def valid_model(epoch):
```

```
# Create figure

fig = plt.figure(figsize=(28, 14))

# setting values to rows and column variablescount=0
psnr = count_psnr(epoch,valid_data=TEST_DIBCO,setting=experiment,flipped=FLIPPED ,
thresh=THRESHOLD)

print('Test PSNR: ', psnr)

#test_images_path=
'C:/Users/Ravinder/Desktop/DocEnTR/DIBCOSETS/'+TEST_DIBCO+'/imgs'

#print('Test Images:', test_images_path)

#output_images_path=
'C:/Users/Ravinder/Desktop/DocEnTR/visbase_256_8/epoch_testing/00_reconstr_no rmal'

#print('Output Images:', output_images_path)

for img in
glob.glob("C:/Users/Ravinder/Desktop/DocEnTR/visbase_256_8/epoch_testing/00_r
econstr_normal/*.png"):

cv_img = cv2.imread(img)B,G,R = cv_img[100,100]
result= int(B)+int(G)+int(R)if result>0:
count=count+1#print(count)
for x in range(count):
name=str(x+1)inputimgadd =

'C:/Users/Ravinder/Desktop/DocEnTR/DIBCOSETS/'+TEST_DIBCO+'/imgs/'+nam e+'.png'

inputimage = cv2.imread(inputimgadd)

outputimgadd =
'C:/Users/Ravinder/Desktop/DocEnTR/visbase_256_8/epoch_testing/00_reconstr_no
rmal/'+name+'.png'
```

```
outputimage = cv2.imread(outputimgadd)plt.subplot(1, 2, 1)
plt.axis('off')

plt.title("INPUT Image-"+name)plt.imshow(inputimage) plt.subplot(1, 2, 2)
plt.axis('off')

plt.title("OUTPUT Image-"+name)plt.imshow(outputimage) plt.show()
model_name = 'C:/Users/Ravinder/Desktop/DocEnTR/weights/train_model.pt'
model.load_state_dict(torch.load(model_name,map_location ='cpu'))epoch = "_testing"
valid_model(epoch)
```

- **requirements.txt:**

```
einops==0.3.2 matplotlib==3.5.0
numpy==1.21.5 opencv_python==4.5.4.60Pillow==9.0.0 thinc==8.0.13 torch==1.10.0
torchvision==0.11.1 tqdm==4.62.3 vit_pytorch==0.24.3
```

- **config.py:**

```
import argparse class Configs ():
def__init_(self): #C:/Users/Ravinder/Desktop/DocEnTR/data/self.parser =
argparse.ArgumentParser()
self.parser.add_argument('--data_path', type=str,
default='C:/Users/Ravinder/Desktop/DocEnTR/data/', help='specify your data path, better
ending with the "/" ')

self.parser.add_argument('--split_size', type=int, default=256, help= "better be a multiple of 8,
like 128, 256, etc ..")

self.parser.add_argument('--vit_patch_size', type=int, default=16 , help=" betterbe a multiple of
2 like 8, 16 etc ..")

self.parser.add_argument('--vit_model_size', type=str, default='base',
choices=['small', 'base', 'large'])
self.parser.add_argument('--testing_dataset', type=str, default='2018')
```

```
self.parser.add_argument('--validation_dataset', type=str, default='2016')
self.parser.add_argument('--batch_size', type=int, default=8) self.parser.add_argument('--
epochs', type=int, default=151, help= 'the desired

training epochs')
self.parser.add_argument('--model_weights_path', type=str, help= 'the desiredtrained model')
def parse(self):
return self.parser.parse_args()
```

7. SYSTEM TEST

7.1 Basics of Software Testing

Software testing is a vital step in the software development cycle. Itensures that the software meets its requirements and functions as expected.

Software testing can be categorized into two essential steps:

1. **Verification**: This step ensures that the software correctly implements aspecific function. It answers the question, “Are we building the product,right?”
2. **Validation**: In contrast, validation ensures that the software aligns with customer requirements. It addresses the question, “Are we buildingthe right product?”

7.1.1 Types of Software Testing

1. Black Box Testing: Black box testing is a software testing technique thatfocusses on a system's functional requirements without exploring into itsinternal mechanisms. Testers do not require knowledge of the system's internal structure; their main goal is to validate functionality based on specified requirements. This method, also known as specification-based testing, assesses an application's functionality without inspecting itsinternal workings. Testers focus on input and output validation rather than the source code.

Types of Black Box Testing:

1. **Functional Testing:** Ensures that every function within the software application operates in position with the specified requirements and standards by validating user interfaces, APIs, databases, security measures, and other components.

2. **Regression Testing:** Verifies that changes made to the code do not introduce fresh errors and confirms compatibility between the existing code and any new additions.

3. **Non-functional Testing (NFT):** Checks non-functional aspects like performance, usability, and scalability.

Advantages of Black Box Testing:

- Independence
- Focus on User Perspective
- Effective for Validation

2. White Box Testing: White box testing, also referred to as clear box testing, glass box testing, or structural testing, is a software testingapproach that entails scrutinizing the internal structure, design, and codeof a software application. In contrast to black box testing, where functionality is tested without insight into internal operations, white boxtesting grants testers access to the source code to create test cases for validating software accuracy at the code level. Testers in white box testing have visibility into the application's source code, enabling them todesign comprehensive test cases that thoroughly evaluate the software from within.

Techniques for White Box Testing:

1. **Statement Coverage:** Ensures that each line of code is executed at least once.
2. **Branch Coverage:** Tests all branches from decision points.
3. **Condition Coverage:** Covers all individual conditions within code.
4. **Multiple Condition Coverage:** Tests all possible combinations ofcondition outcomes.
5. **Basis Path Testing:** Constructs control flow graphs and calculatesCyclomatic complexity to design minimal test cases for independent paths.

Advantages of White Box Testing:

- Early issue detection.
- Thorough code coverage.

Black Box Testing	White Box testing
Tester will perform testing without having knowledge of the internal structure	Tester will perform testing with strong knowledge of the internal structure
Higher level testing to check the application behaviour and validate the whole system from the user's perspective	Lower level testing focusing on the application's internal operation to analyze code structure (Unit testing, Integration testing)
Can be done by the testers having no programming knowledge	Require high-skilled testers or developers with good programming knowledge
Provide simple test reports	Provide detailed reports
It has lower complexity, a swift process	Involves complicated steps, time-consuming process

Fig. 7.1 Black Box and White Box Testing

7.1.2 Levels of Software Testing

There are 4 different levels of testing. They are Unit Testing, IntegrationTesting, System Testing, and Acceptance Testing.

1. Unit Testing

Unit testing serves as the initial level of software testing. Its purpose is to verify whether software modules meet the specified requirements. It focuses on individual components or units of a softwareapplication. These units can be functions, methods, or other small parts of the code. Unit testing is also the first level of functional testing. It ensures that each unit performs as intended. A unit component representsthe smallest testable part of the software. It could be an individual function or a specific regulation within the application. The primary goalof unit testing is to validate the correctness of these isolated code units.

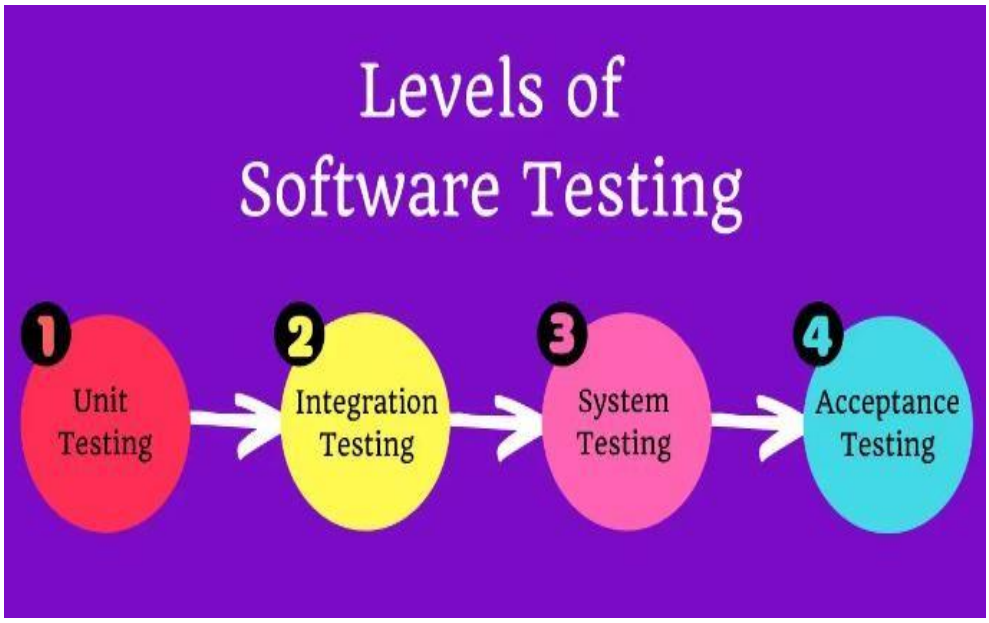

Fig. 7.2 Levels of Software Testing

2. Integration Testing

Integration Testing is a crucial step in the software testing process. Integration testing occurs after unit testing and focuses on verifying the interactions and data exchange between different components or modulesof a software application. The primary goal of integration testing is to expose faults in the interaction between integrated units. When each component or module works independently, integration testing checks thedata flow between dependent modules. It is performed once functional testing has successfully validated each application module. It evaluates theaccuracy of communication among all the modules.

3. System Testing

System Testing represents the third level of software testing. System testing involves evaluating the entire software system. It verifies both functional and non-functional requirements. In system testing, we assess the application's end-to-end flow, simulating how a user interacts with the software. The testing environment closely resembles the production environment. It navigates through all necessary modules of anapplication, ensuring that end features and business processes function correctly. It examines the complete working of an integrated software system against specified requirements.

4. Acceptance Testing

Acceptance Testing represents the fourth and final level of softwaretesting. Acceptance testing evaluates whether the software conforms to specified requirements and meets user expectations. Even after unit testing, integration testing, and system testing, minor errors may surface when the system is used by end-users in real-world scenarios. User Acceptance Testing (UAT), acceptance testing is performed by the customer before accepting the final product. UAT concentrates on featuresand scenarios regularly used by customers or end-users.

7.2 Functional and Non-Functional Testing

Fig. 7.3 Functional and Non-Functional Testing

7.2.1 Functional Testing

Functional testing is a software testing approach that evaluates whether a system adheres to its functional requirements and specifications. It ensures that the application satisfies the specified requirements. This type of testing focuses on the system's behavior during processing, simulating real-world usage scenarios. Unlike non-functional testing, it does not delve into system architecture assumptions or source code. Each function of the software application is tested by providing relevant input, comparing the actual output withthe expected output.

7.2.2 Non-Functional Testing

Non-functional testing verifies aspects of a software application beyond its functional requirements. It examines whether the system behaves as expectedin terms of non-functional parameters. Non-functional testing covers areas not addressed by functional testing. Parameters such as performance, scalability, security, and usability fall under non-functional testing. Ensuring non-functionalreadiness is equally crucial as functional testing for a robust system.

8. OUTPUT SCREENS

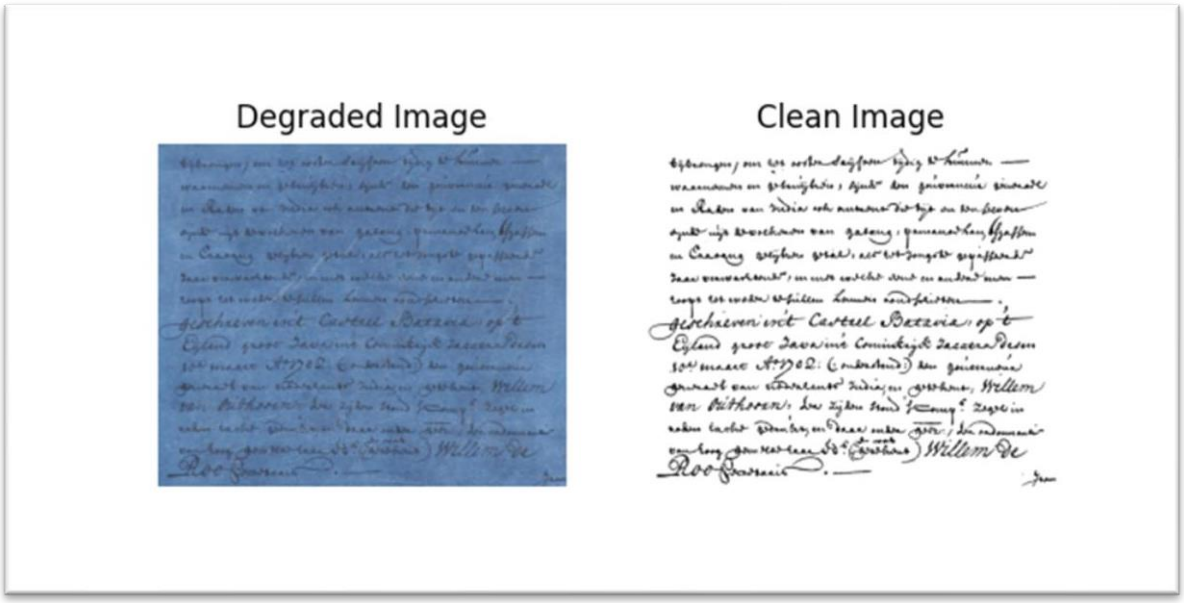

Fig. 8.1 Enhanced image 1

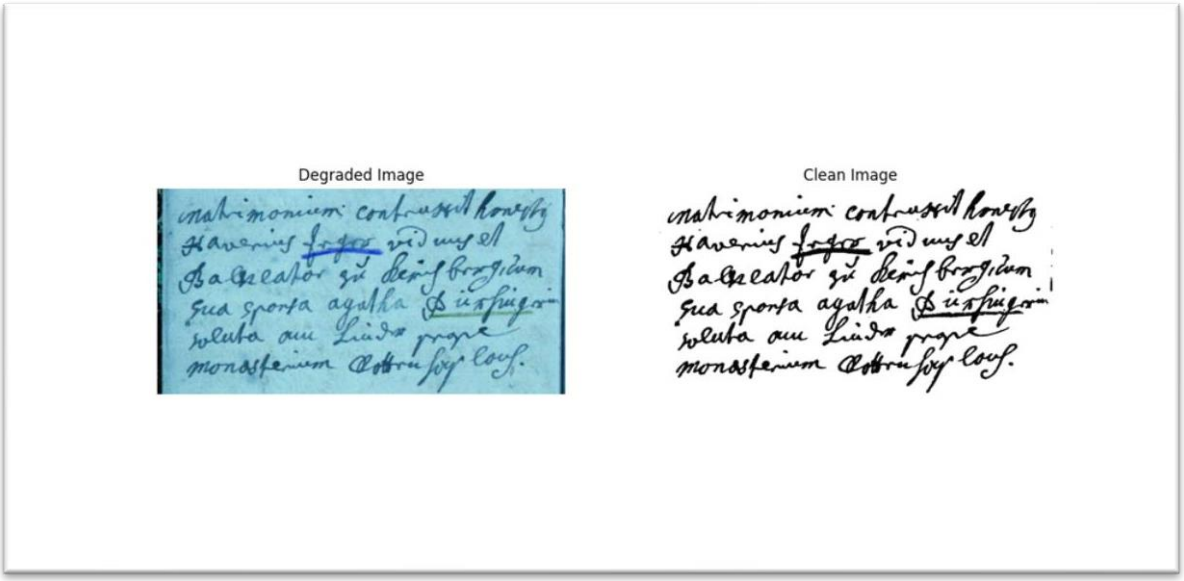

Fig. 8.2 Enhanced image 2

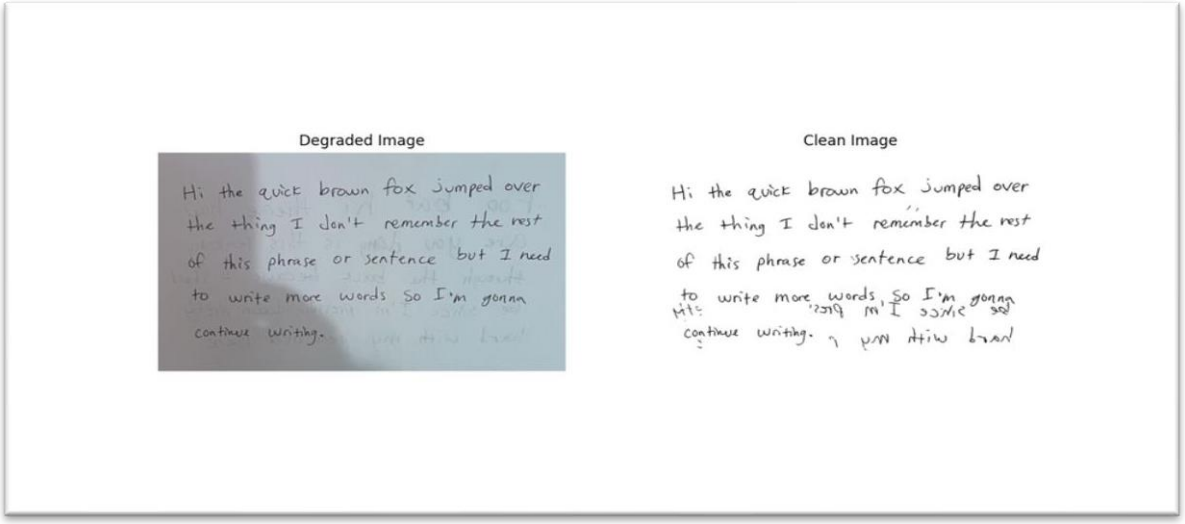

Fig. 8.3 Enhanced image 3

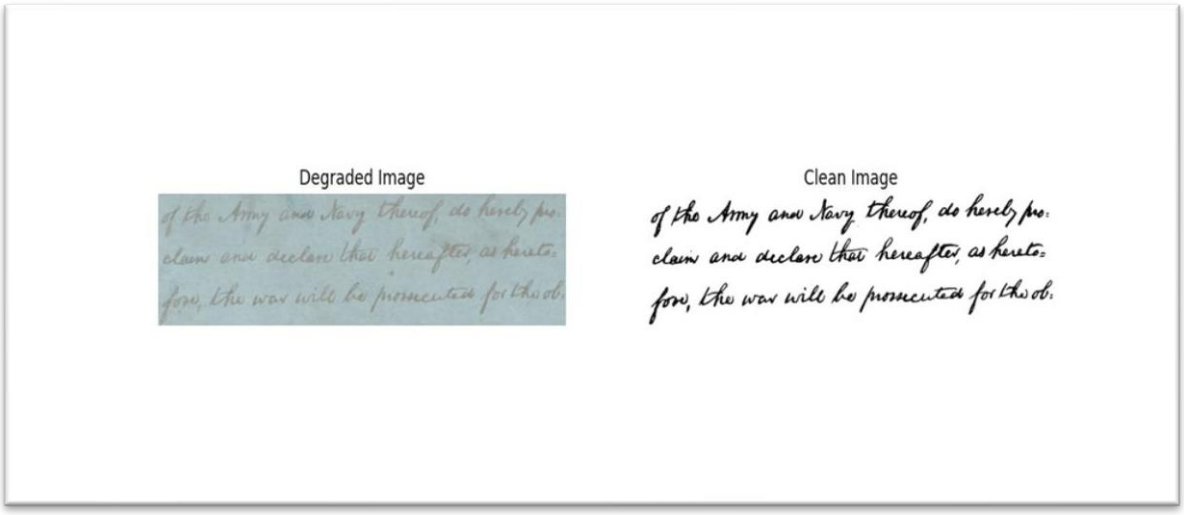

Fig. 8.4 Enhanced image 4

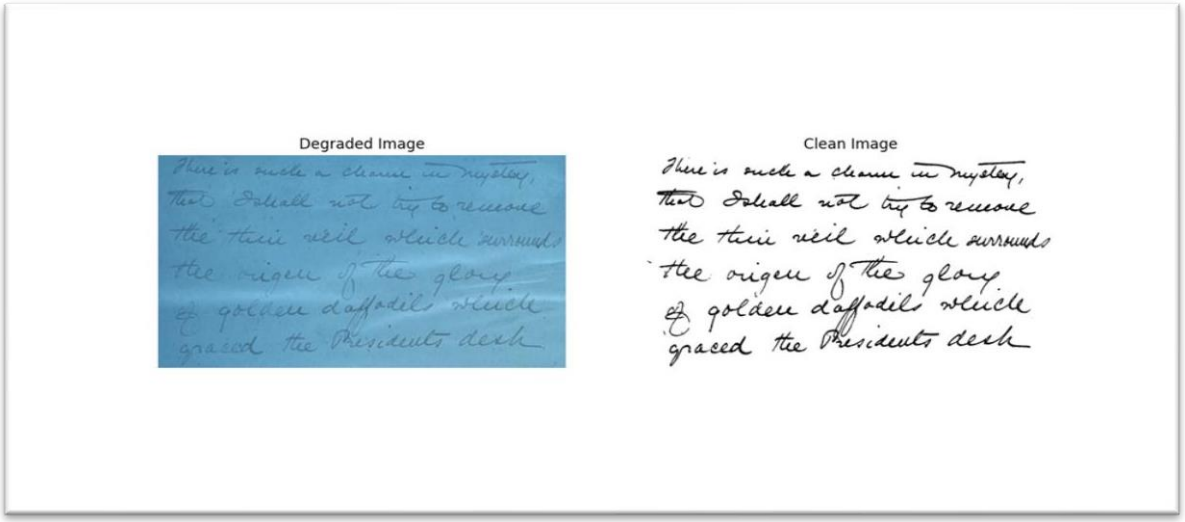

Fig. 8.5 Enhanced image 5

Fig. 8.6 Enhanced image 6

9. CONCLUSION

The study of Document Image Enhancement using Vision Transformers introducesan innovative transformer-based architecture designed for enhancing document images. This architecture leverages self-attention mechanisms too capture extensive global dependencies hence enhancing the performance. To the best of our knowledge, this is the first pure transformer model addressing DIE related problems. The model captures high level global long-range dependencies using the self-attention mechanism for a better performance. Quantitative and qualitative results on the DIBCO benchmarks prove the effectiveness of this project in recovering highly degraded document images. It is a simpleand flexible framework that can also be easily applied to enhance other kinds of degradation occurring in document images (like blur, shadow, warps, stains etc).

10. FUTURE ENHANCEMENTS

Some of the new aspects will be investigated in a future work. We also wish to investigate a self-supervised learning stage that can substantially benefit from large amounts of unlabelled data. We can also optimize the computational efficiency of visiontransformers to reduce training and inference time, making the process cost effective and more scalable.

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DOCUMENT IMAGE ENHANCEMENT USING VISION TRANSFORMERS

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ABSTRACT:

Document images can undergo numerous degradation scenarios, causing difficulties in recognition and processing. In the digital era, it is crucial to remove noise from these images to ensure their proper utilization. To tackle this issue, a new encoder-decoder architecture based on vision transformers has been developed to improve both machine-printed and handwritten document images seamlessly. The encoder works directly on pixel patches along with their positional data, eliminating the need for convolutional layers, and the decoder recreates a clean image from the encoded patches. Experimental results indicate the superiority of this model over existing methods on various DIBCO benchmarks.

Keywords: Degraded images, Encoder-Decoder Architecture, DIBCO

I INTRODUCTION

Preserving the legibility and quality of document images, especially historical ones, is a critical focus within Document Image Analysis and Recognition (DIAR) research. Historical documents often hold valuable information spanning centuries and decades. However, these documents are susceptible to various forms of degradation such as smears, stains, pen strokes, and uneven illumination, which can hinder their preservation.

The presence of such distortions can significantly impact downstream tasks in information processing, including segmentation, Optical Character Recognition (OCR), information spotting, and layout analysis. To address these challenges, a robust pre-processing step is

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essential to denoise and reconstruct degraded images into high-quality, clean versions. Document Image Enhancement (DIE) plays a crucial role in restoring the quality of degraded document samples to produce clear, uniform images.

Recent advancements in Convolutional Neural Network (CNN)-based approaches have revolutionized DIE tasks such as binarization, deblurring, shadow removal, and watermark removal. While CNNs have shown improved performance over traditional methods, they have limitations. CNNs may not be optimal for restoring different regions of a document image due to their grid-based operation and limited ability to capture long-range dependencies.

In response to these limitations, Vision Transformers (ViTs) have emerged as a promising alternative. ViTs split images into fixed-size patches, enabling the capture of both local and global spatial dependencies. By leveraging the self-attention mechanism inspired by transformers in Natural Language Processing (NLP), ViTs excel at capturing global interactions and contextual features in images.

Our proposed Document Image Enhancement using Vision Transformer leverages ViTs in an encoder-decoder framework, eliminating the need for CNNs. This aims to restore and enhance degraded document images efficiently. By incorporating ViTs and self-attention mechanisms, it can recover missing or degraded patches by reasoning globally between neighbouring patches.

The innovative approach of this project introduces a transformer-based model for document image enhancement, showcasing the power of ViTs.

II. LITERATURE SURVEY

1. Transformers in Vision: A Survey - arXiv

It discusses image restoration Transformer models for various tasks like image super-resolution, image enhancement, and colorization. It compares advantages and limitations of popular techniques in computer vision. It highlights the applications of transformers in image processing tasks and provides insights into future research directions.

2. An Overview of Vision Transformers for Image Processing: A Survey - IJACSA

It focuses on the success of vision transformer-based models in artificial intelligence, particularly in computer vision. It emphasizes the potential applications of vision transformers for student verification in university systems. It discusses the importance of accurately

verifying student identities for academic success in online learning environments.

3. A Survey on Visual Transformer - arXiv

It provides a complete overview of transformer-based models in computer vision, covering object detection, image classification, and video processing. It discusses the applications of transformers in high/mid-level vision, low-level vision, and video processing tasks. It highlights the performance of transformer models in various visual tasks and their potential for improving a wide range of visual applications.

4. Vision Transformers in Image Restoration: A Survey - MDPI

It surveys the impact of Vision Transformers in image restoration tasks like image super-resolution, denoising, and deblurring. It compares the efficiency metrics of ViT-based models on different benchmarks and datasets for image restoration. It discusses the challenges and future work in using Vision Transformers for image restoration tasks.

III SYSTEM ANALYSIS

EXISTING SYSTEM

The literature suggests using Convolutional Neural Networks to predict and classify four types of skin lesions. A website is created for real-time use of the model, which predicts the three most common categories of skin lesions for a certain images. The experiment was conducted using the MNIST: HAM10000 dataset, which includes 100,000 annotated images. Another researcher developed an algorithm to detect skin lesions early by extracting features using the ABCD rule, GLCM, and HOG. Preprocessing is used to improve skin lesions quality and clarity are used to reduce artifacts such as skin color, hair, and so on. Geodesic Active Contour (GAC) was used to segment the lesion, allowing for distinct feature extraction. The ABCD scoring system was utilized to extract attributes like symmetry, border, color, and diameter. HOG and GLCM were used to extract textural information. The retrieved features are used to categorize skin lesions as benign or melanoma using machine learning algorithms including SVM, KNN, and Naïve Bayes. The project utilized skin lesion images from the International Skin imaging Collaboration, including 328 benign images and 672 melanoma images.

Limitations of Existing system

1. Grid-Based Operation
2. Limited Capture of High-Level Dependencies

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3. Task-Specific Filters
4. Less Effective for Long-Range Dependencies

PROPOSED SYSTEM

In this proposed system, a simple Document Image Enhancement Transformer is used. It is an end-to-end approach that effectively restores and improves degraded document images. It utilizes Vision Transformers in an encoder-decoder framework. Document Image Binarization Competition (DIBCO) shows that this achieves state-of-the-art results for both machine-printed and handwritten degraded document images

Proposed system Advantages:

1. Transformer-Based Architecture
2. Encoder-Decoder Framework
3. Flexibility and Simplicity

IV IMPLEMENTATION

Architecture:

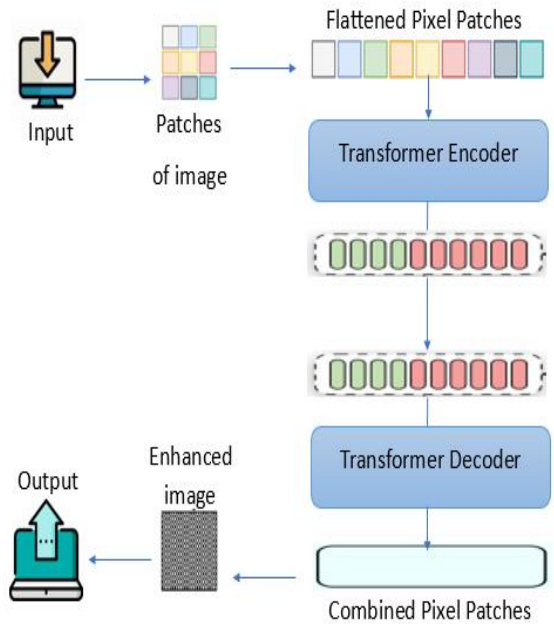

Fig-1. Architectures of the system model

1. System Architecture

The proposed system architecture consists of the encoder-decoder framework which works on the basis of Vision Transformers.

- **Input Image:** Degraded Image is taken as a input, it can be handwritten or printed.
- **Image Patches:** The input image is divided into smaller pixel patches; each patch contains a subset of pixels from the original image.
- **Flattened Pixel Patches:** The patches are then divided into a flattened pixel patch which represents the data in one-dimensional format, making the process further easy.

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- **Transformer Encoder:** This uses the Vision Transformers architecture and encodes the data from the flattened pixel patches using the positional information.
- **Transformer Decoder:** This decodes the encoded data by reconstructing meaningful patterns, and uses the self-attention mechanisms for context-aware decoding.
- **Combined Pixel Patches:** The decoded data is constructed into the combined pixel patch which represents the segments of the enhanced image.
- **Enhanced Output:** This assembles all the combined pixel patches to form a complete enhanced image by improved quality, visual appeal, colour correction or other features.

2. Modules

In this paper, there are 2 modules based on the architecture of Vision Transformers.

2.1 Encoder

The initial step involves partitioning an image into a collection of patches. These patches are then transformed into tokens, incorporating their positional data. Subsequently, a series of transformer blocks is utilized to convert these tokens into the encoded latent representation. These blocks adhere to a structure similar to, comprising layers of multi-headed self-attention and multi-layered perceptron (MLP). Each block is prefaced by a LayerNorm (LN) and succeeded by a residual connection. The dimensions of the patches' embedding and the quantity of transformer blocks are determined based on the model's size.

- Initial encoding stage divides image into patches
- Patches embedded to create tokens with positional information
- Transformer blocks used to map tokens into encoded latent representation
- Blocks follow structure described in reference: multi-headed self-attention and multi-layered perceptron layers

2.2. Decoder

The decoder segment comprises a sequence of transformer blocks (equivalent in number to the encoder blocks) that receive the outputted tokens from the encoder as input. These tokens progress through the transformer decoder blocks and are then projected using a linear layer to generate the desired pixel values. This configuration ensures that each output element corresponds to a vector representing a flattened patch in the resultant image. Ground truth pixel values are derived by segmenting the ground truth

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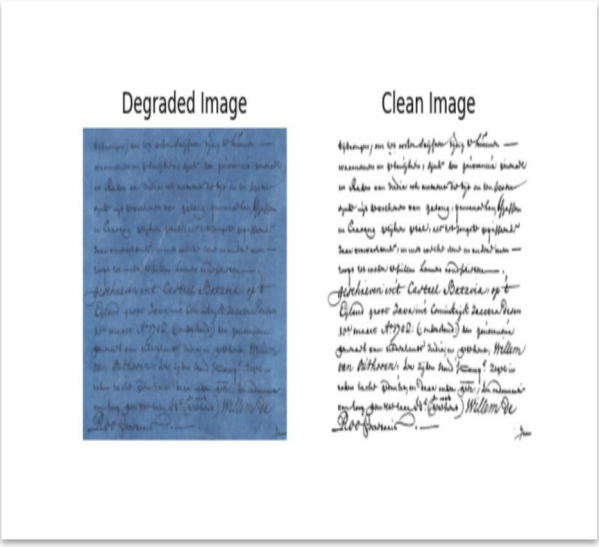
(GT) clean image into patches (following the same approach as with the input degraded image) and flattening them into vectors. Training the model involves utilizing a mean squared error (MSE) loss function to compare the model's output with the GT pixel patches.

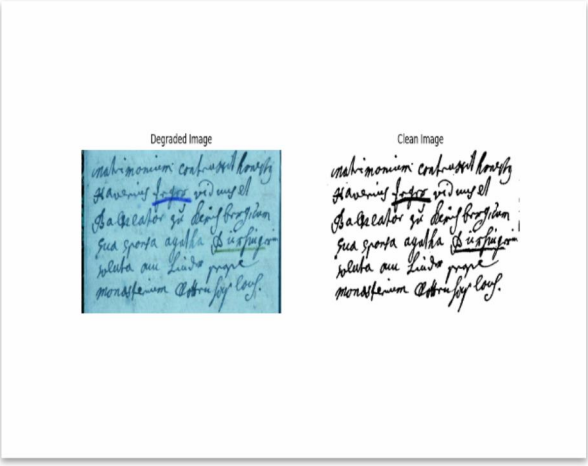
- Decoder component consists of transformer blocks, matching encoder's number
- Takes encoder's token sequence as input
- Tokens processed within transformer decoder blocks
- Projected using linear layer to get pixel values
- Each output element represents flattened patch in resulting image

V RESULT AND DISCUSSION

Enhanced image 1:



Enhanced image 2:



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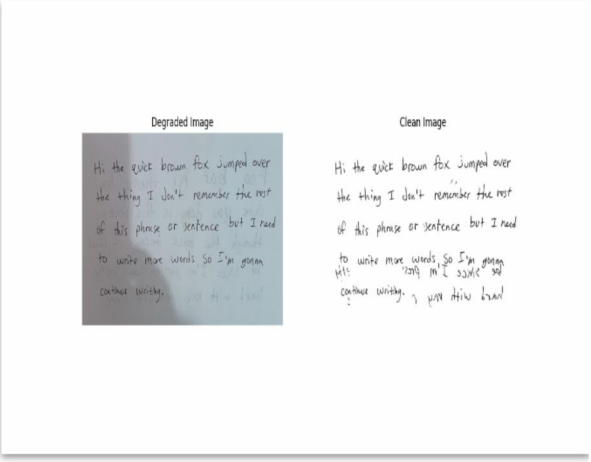
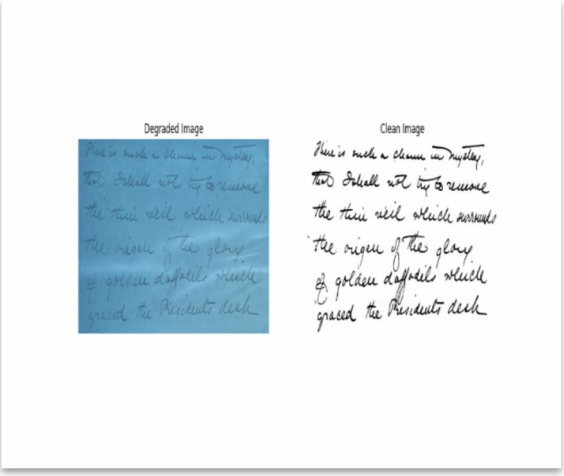
Enhanced image 3:



Enhanced image 4:

Enhanced image 5:



Enhanced image 6:

VI CONCLUSION

The study of Document Image Enhancement using Vision Transformers introduces an innovative transformer-based architecture designed for enhancing document images. This architecture leverages self-attention mechanisms to capture extensive global dependencies hence enhancing the performance. To the best of our knowledge, this is the first pure transformer model addressing DIE related problems. The model captures high level global long-range dependencies using the self-attention mechanism for a better performance. Quantitative and qualitative results on the DIBCO benchmarks prove the effectiveness of this project in recovering highly degraded document images. It is a simple and flexible framework that can also be easily applied to enhance other kinds of degradation occurring in document images (like blur, shadow, warps, stains etc.).

FUTURE ENHANCEMENTS

Some of the new aspects will be investigated in a future work. We also wish to investigate a self-supervised learning stage that can substantially benefit from large amounts of unlabelled data. We can also optimize the computational efficiency of vision transformers to reduce training and inference time, making the process cost effective and more scalable.

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